Reminders and announcements

- During lectures you can submit questions to: vevox.app ID: 107-018-952
- You can find answers to questions in the COMP1870 Teams Channel (link in Minerva) (along with other discussion).
- Maths Lunch Hours, Mondays and Wednesdays, 1.00pm - 2.00pm. Bragg GR.18.

Revision

- A **set** is a collection of objects, which are called **elements** of that set.
- $x \in A$ means that x is an element of the set A.
- Sets of numbers:
  - The **integers**: $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$.
  - The **rational numbers**: $\mathbb{Q} = \{p/q \mid p, q \in \mathbb{Z},\ q \neq 0\}$.
  - The **real numbers** $\mathbb{R}$: the set of numbers with decimal representations.
- $\emptyset$ is the **empty set**, the unique set with no elements.
- $A \subseteq B$ means that A is a **subset** of B; that is, every element of A is an element of B.
- $A \subset B$ means that A is a **proper subset** of B; that is, A is a subset of B and $A \neq B$.
- If A is a set then $\mathcal{P}(A)$ is the **power set** of A; that is, the set containing all subsets of A.

Ordered pairs

An **ordered pair** is a pair (a, b) where a and b are elements.

Order matters and repetition is allowed. We use round brackets instead of curly brackets to emphasise the difference between an ordered pair and a set.

Cartesian products

Assume A and B are sets. Then $A \times B$ is the **Cartesian product** of A and B. It is the set of all ordered pairs with the first element from A and the second element from B.

$$A \times B = \{(a, b) \mid a \in A,\ b \in B\}$$

Tuples

An ***n*-tuple** is an ordered sequence of n elements, written as $(a_1, a_2, \ldots, a_n)$.

If $A_1, A_2, \ldots, A_n$ are sets, then $A_1 \times A_2 \times \cdots \times A_n$ is the set of all tuples where the element in position i is from A_i.

$$A_1 \times A_2 \times \cdots \times A_n = \{(a_1, a_2, \ldots, a_n) \mid a_1 \in A_1, a_2 \in A_2, a_3 \in A_3, \ldots, a_n \in A_n\}$$

Data structures

A **data structure** is some method of storing information so that a computer can access it. Choosing the correct data structure is important to make algorithms run efficiently.

Sets and tuples are common data structures.

A set is an unordered data structure, and a **list** is an ordered structure (basically the same as a tuple).

Operations on data structures

Operations on sets:

- `is_empty(S)`
- `is_element_of(x,S)`
- `add(x,S)`
- `remove(x,S)`
- `union(S,T)`
- `intersection(S,T)`

Operations on lists:

- `append(x,S)`
- `prepend(x,S)`

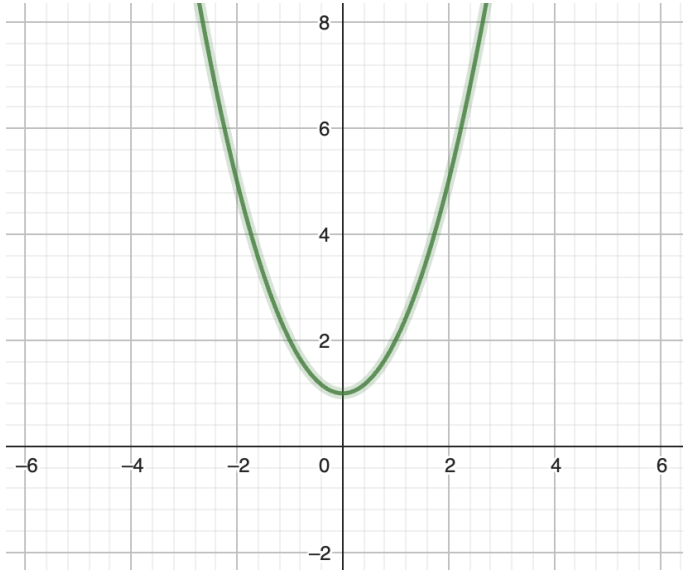
Functions

The idea of a **function** is central to mathematics and computer science.

We can think of a function as being a machine which takes input and produces output. Each possible input should produce exactly one output.

Functions

The functions we are most familiar with take a real number as input and produce another real number as output.



Functions

In general, there doesn't need to be a nice rule like $y = x^2 + 1$ which turns inputs to outputs. Functions can be arbitrary as long as each input corresponds to exactly one output.

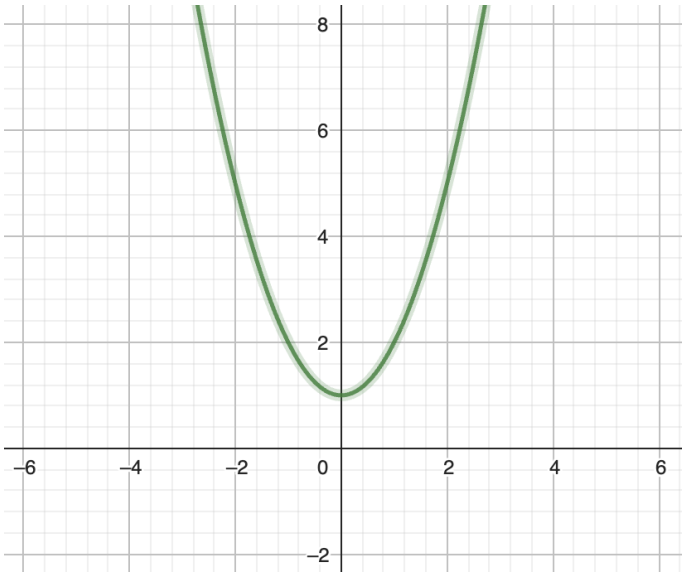
The set of inputs is called the **domain**. The set of potential outputs is called the **codomain**.

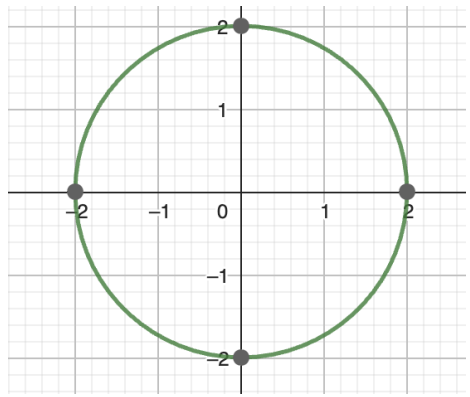
The notation $f : A \to B$ means that f is the name of a function with A as its domain and B as its codomain.

If the input $a \in B$ produces the output $b \in B$, we say a is **mapped** to b, and we write $f(a) = b$ or $f : a \mapsto b$.

Functions

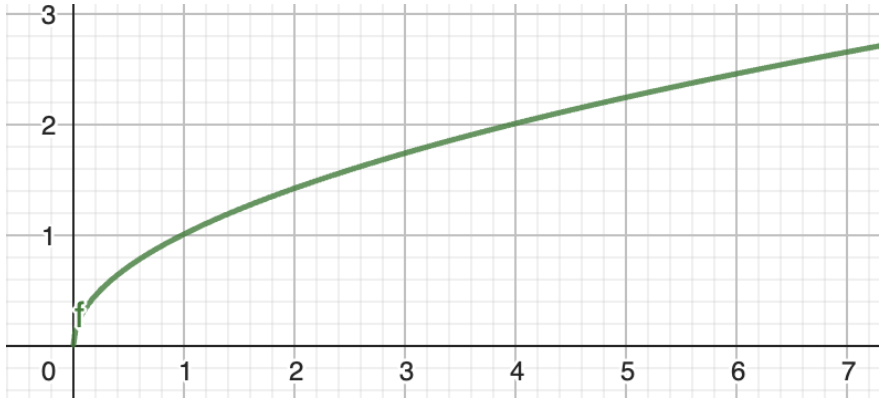
If $f : \mathbb{R} \to \mathbb{R}$ is a function, then any vertical line crosses the graph of f at most once. This is because each input has exactly one output. This is called the **vertical line test**.





Functions

If we use the convention that $\sqrt{x}$ is greater than or equal to zero whenever x is a non-negative real number, then there is a function with domain and codomain equal to the non-negative real numbers, taking x to $\sqrt{x}$.



Functions

The codomain is just the set of **potential** outputs. Some elements of the codomain may never appear as outputs.

The set of all outputs is called the **image** (or **range**) of the function.

If $f : A \to B$ is a function, its image is

$$\{f(a) \mid a \in A\}.$$

This is a subset of the codomain B, possibly a proper subset.

Functions

We think of a function as a set of ordered pairs.

If $f : A \to B$ is a function, then the function contains the ordered pair (a, b) if f maps a to b.

In other words, f is a subset of the Cartesian product $A \times B$.

Functions

A function can't be an arbitrary subset of $A \times B$, because we require that every input produces exactly one output.

Definition 6.1. Assume that A and B are sets. A **function** with A as its domain and B as its codomain is a subset $f \subseteq A \times B$ such that for every element $a \in A$, there is exactly one ordered pair in f with a as its first element.

Arrow diagrams

If $f : A \to B$ is a function, we may be able to represent it by drawing a diagram showing arrows from elements of A to elements of B. Each arrow represents an ordered pair in the function.

Composition of functions

We are familiar with chaining together functions, as in $y = (x+1)^2$. This process is called **composition** of functions.

Composition of functions

Say that $f : A \to B$ and $g : B \to C$ are functions. Imagine the function that takes an element of a and operates on it with f and then g. The resulting function is denoted by $g \circ f$.

This function takes each element $a \in A$ to $g(f(a))$.

Properties of functions

Definition 6.4. A function $f : A \to B$ is **one-to-one** or **injective** if $f(a_1) = f(a_2)$ implies $a_1 = a_2$.

In the contrapositive: $a_1 \neq a_2$ implies $f(a_1) \neq f(a_2)$.

This means different arrows go to different locations.

Properties of functions

Definition 6.5. A function $f : A \to B$ is **onto** or **surjective** if for every element $b \in B$, there is an element $a \in A$ such that $f(a) = b$.

This means that every element of the codomain has an arrow going to it.

Properties of functions

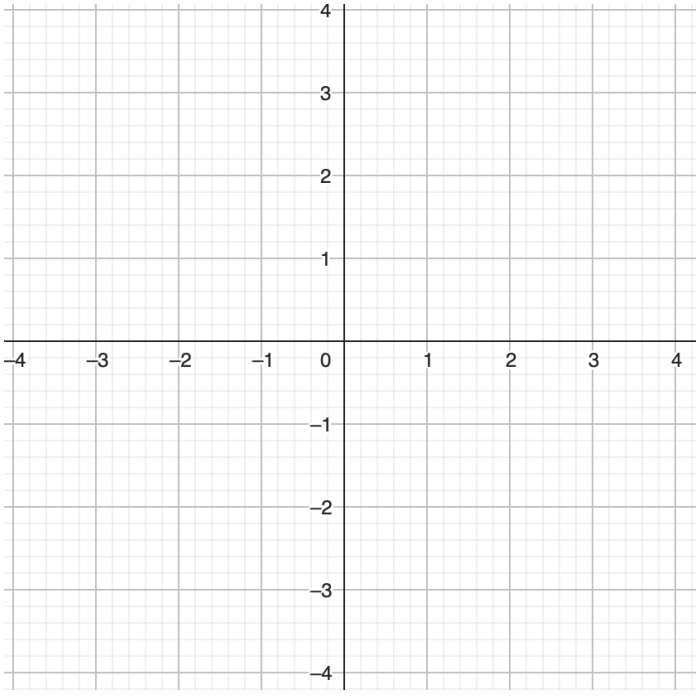
A function $f : A \to B$ is **bijective** if it is injective and surjective.

Properties of functions

Property	**Definition**	**Characterisation via arrows**	**Characterisation via ordered pairs**
$f : A \to B$ is a function.	Each input is mapped to exactly one output.	Each element of A has exactly one arrow coming out out of it.	For every $a \in A$ there is exactly one ordered pair in f with a as the first element.
$f : A \to B$ is injective.	$f(a_1) = f(a_2)$ implies $a_1 = a_2$.	Different arrows go to different locations.	If (a_1, b) and (a_2, b) both belong to f then $a_1 = a_2$.
$f : A \to B$ is surjective.	The image of f is equal to B.	Every element of B has an arrow pointing to it.	For every $b \in B$ there exists an $a \in A$ such that (a, b) belongs to f.

Identities and inverses

Let A be an arbitrary set. The **identity function** on A maps every element of A to itself. It is denoted by id_A.



Identities and inverses

Let $f : A \to B$ be a function.

An **inverse** of f is a function $f^{-1} : B \to A$ such that

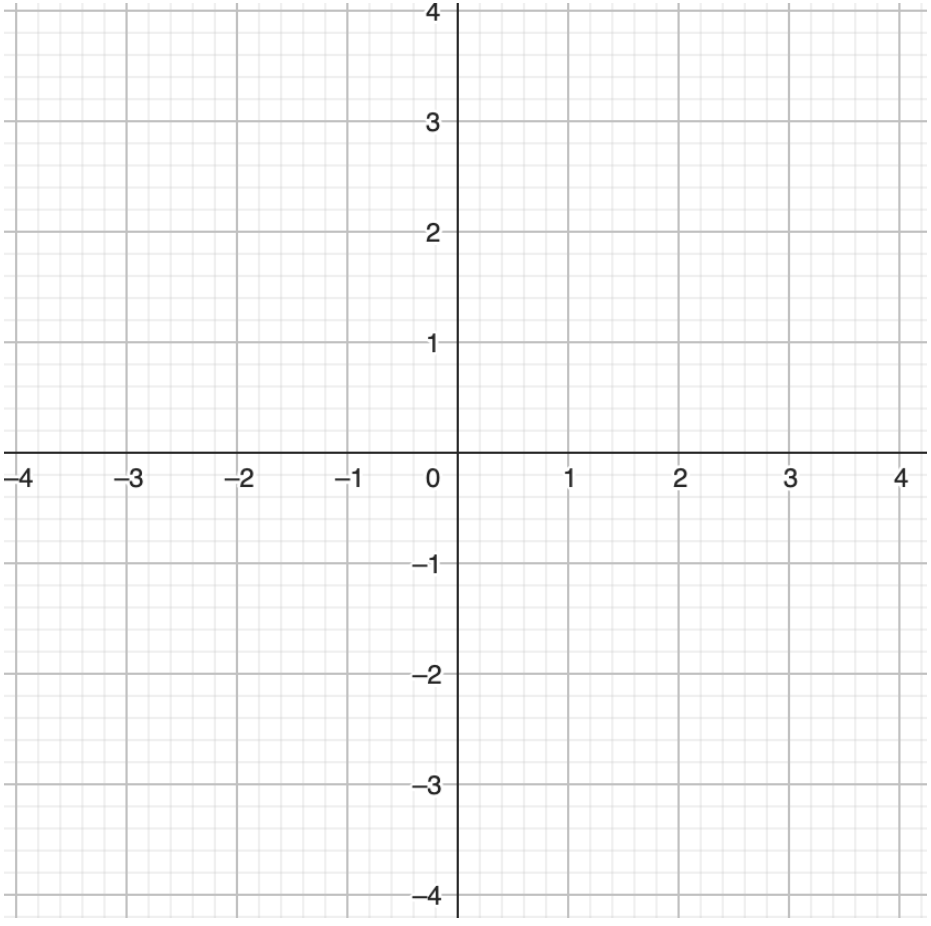
$$f^{-1} \circ f = \mathrm{id}_A \qquad \text{and} \qquad f \circ f^{-1} = \mathrm{id}_B$$

Not every function has an inverse.

If f has an inverse, then it is obtained by "flipping" every ordered pair in f.

$$f^{-1} = \{(b, a) \mid (a, b) \in f\}$$

When does a function have an inverse?



When does a function have an inverse?

When does a function have an inverse?

Theorem 6.9. Let $f : A \to B$ be a function. Then f has an inverse if and only if it is a bijection.